Aadit Kshirsagar

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EDUCATION

University of Southern California — Aug 2025 - Expected May 2027
Master of Science in Computer Science – Artificial Intelligence

Savitribai Phule Pune University — Aug 2020 - Jun 2024
Bachelor of Engineering in Computer Science — GPA:8.89/10

WORK EXPERIENCE

Calsoft Pvt Ltd | Data Science Intern | Pune, India — Apr 2024 - Jun 2025

- Key member of the core GenAI team, leading end-to-end project execution, from architecture to optimization, delivering scalable, AI-driven solutions.
- Built a multi-pipeline, agent-based architecture that increased test case relevance accuracy to 95% and enabled retrieval of legacy test cases.
- Designed and implemented an on-premises AI solution using open-source LLM's to automate test case generation reducing manual effort by 80% and increasing QA productivity.

Intellect Design Arena Ltd | AI/ML Intern | Pune, India — Sep 2023 - Mar 2024

- Played a key role in the development of iGTB Copilot, an internal project. The Copilot was designed to enable Relationship Managers to leverage actionable insights from bank data.
- Contributed to LLM integration and prompt engineering for a successful proof of concept of a Copilot assisting with payment onboarding.

Intellect Design Arena Ltd | Software Developer Intern | Pune, India — Jan 2023 - Mar 2023

- Implemented various Natural Language Processing techniques to carry out efficient filtering of bank and customer data.
- Successfully executed a project utilizing Streamlit and AI technologies, resulting in a 45% reduction in Know Your Customer checks for bank customers.

PROJECTS

Text Gen from Scratch | Codebase

- Designed and implemented a transformer-based text generation model from scratch, featuring multi-head self-attention, positional encoding, and a custom learning rate scheduler.
- Developed an end-to-end training pipeline with custom vocabulary construction, efficient tokenization, and temperature-based sampling for controllable, high-quality autoregressive text generation.

Medi Match Copilot | Codebase

- Integrated llama3.1 for natural language processing and llama3.2-vision for medical image analysis, enabling robust multi-modal reasoning in clinical workflows.
- Architected a modular Python-based system for scalable retrieval of medical data, generation of diagnostic hypotheses, and visual analysis, designed for long-term maintainability.
- Delivered a comprehensive clinical assistant that automates patient symptom interpretation, medical image evaluation, and evidence-based knowledge synthesis, assisting in making data-informed diagnostic decisions.

Eco Policy Copilot | Codebase

- Integrated advanced data modelling and workflow automation to support evidence-based decision-making for climate and economic policy initiatives, assisting environmental researchers and analysts.
- Developed an AI-driven platform using LangChain and LangGraph that analyses climate, economic, and energy data, delivering actionable policy recommendations for sustainable development.
- Designed and implemented an agentic workflow, enabling autonomous agents to collaboratively process data, simulate policies, and generate executive summaries for streamlined environmental analysis.

TECHNICAL SKILLS

Programming: Python, Java, SQL, Git
Databases: PostgreSQL, MongoDB, Chroma, Pinecone, Milvus, Weaviate
Web Technologies: HTML, CSS, JavaScript, Streamlit
Cloud and DevOps: Azure DevOps, Google Cloud, Amazon Web Services, Fast API, Docker, Kubernetes, Azure AI, Open AI, Amazon Bedrock
AI/ML: Generative AI, Machine Learning, Deep Learning, NLP, LLMOps, MLOps, RAG, AutoGen, MCP, LangChain, LlamaIndex, LangGraph, Prompt Engineering, Computer Vison, Transformers, LoRa, Pandas, PySpark, PyTorch, TensorFlow

PUBLICATIONS AND ACHIEVEMENTS

- "Enhancing RAG performance through Chunking and Text Splitting Techniques" published in the International Journal of Scientific Research in Computer Science, Engineering, and Information Technology. | Link
- Secured 3rd place with my team at IIT Bombay's Tech fest, Asia's largest technology competition.